Capstone Project: Secure Business Network Deployment

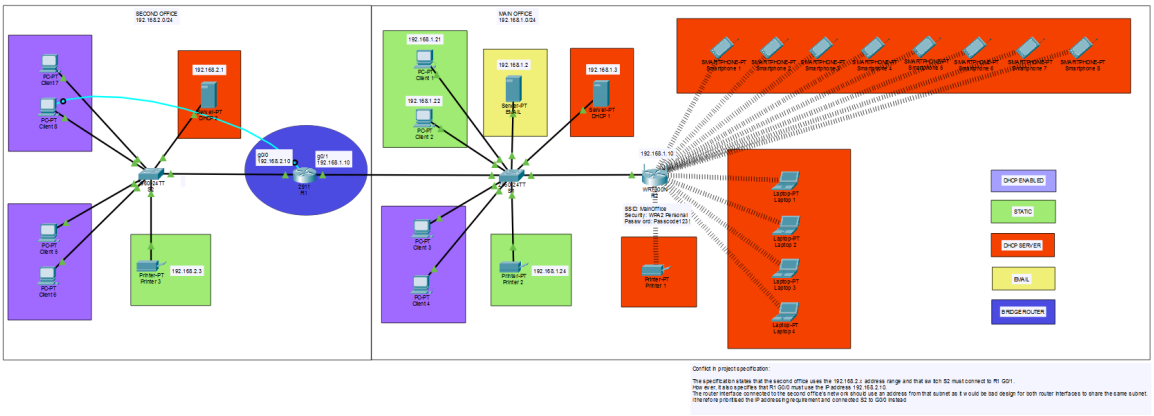
Packet Tracer Practical Assessment | Student Workbook

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Date:	15/10/2026

Scenario

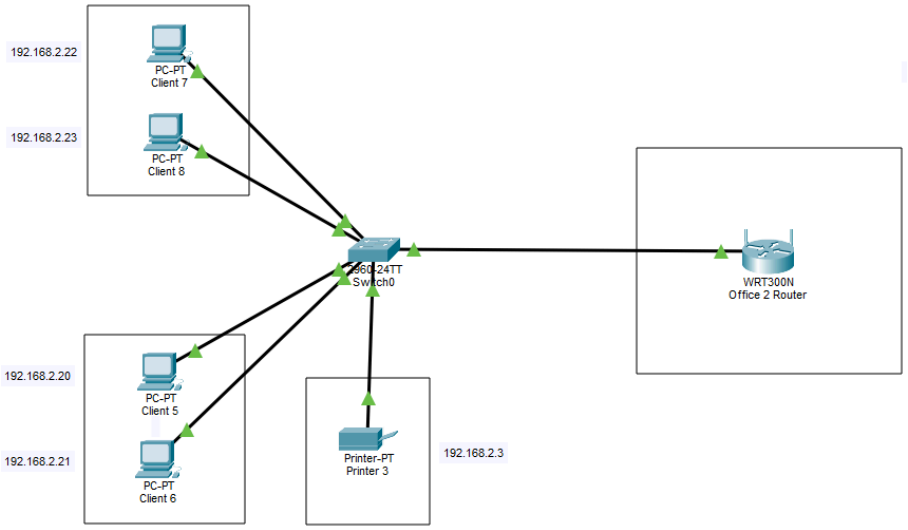
You are a junior network engineer working for a growing company called TechSolutions Ltd. The company has opened additional office spaces and requires a secure and fully functional network infrastructure. You must design, configure, secure, and test the network using Cisco Packet Tracer.

Screenshot of final topology:



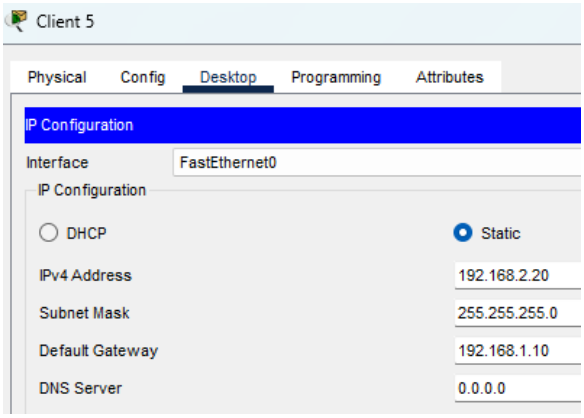
TASK 1 – Basic Network Topology

Paste Screenshot of Full Topology (after Task 1) with correct cabling and device labels here:

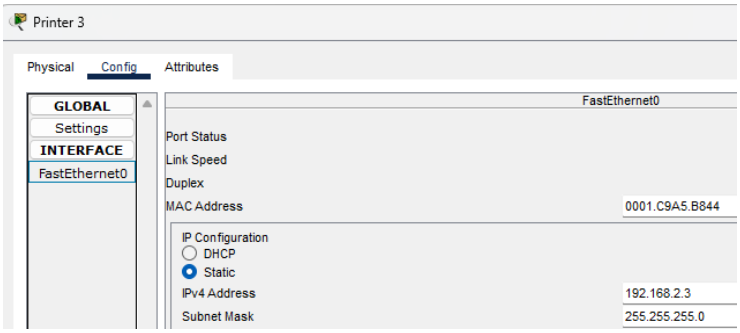


Note - Original version of project specification did not require DHCP setup in Task 1.

Paste Screenshot of the IP configuration of Client 5 PC here:



Paste Screenshot of the IP configuration of Printer 3 here:

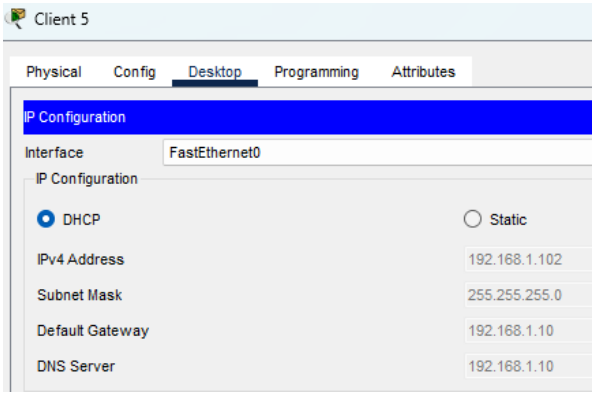


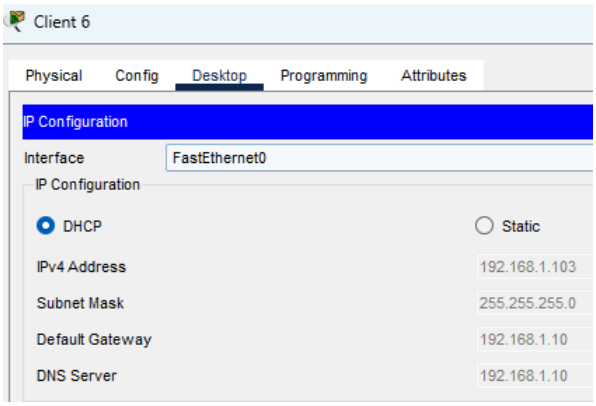
TASK 2 – DHCP Configuration

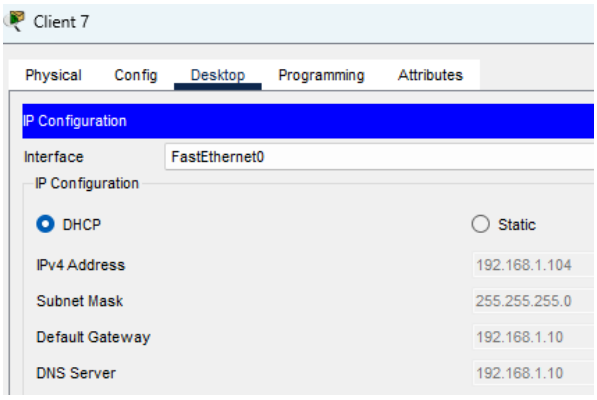
Paste Screenshot of DHCP pool configuration here:

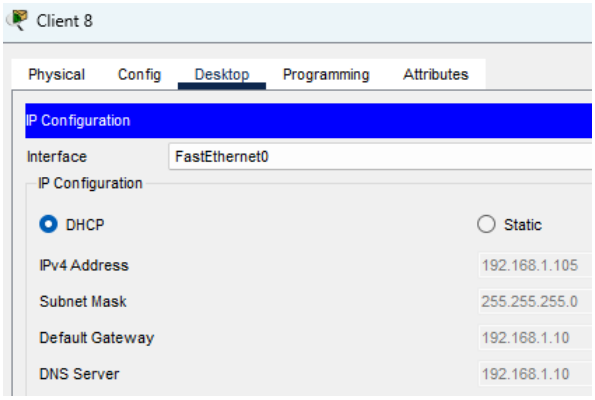
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Office2	192.168.1.10	192.168.1.10	192.168.2.50	255.255.255.0	100	0.0.0.0	0.0.0.0

Paste Screenshots of Clients 5 – 8 IP configuration (Showing DHCP addresses) here;



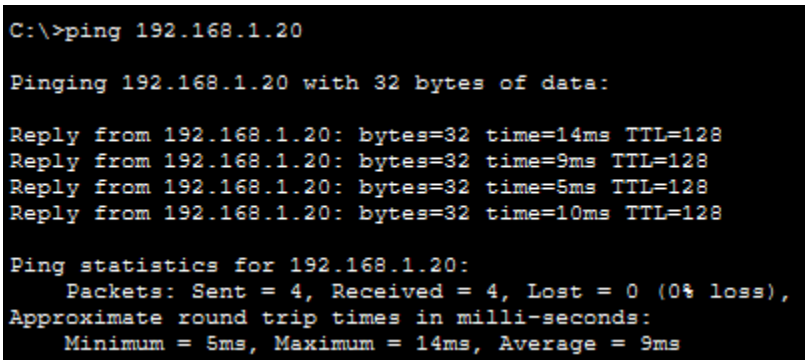

Client 6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.103
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.10
DNS Server
192.168.1.10


Client 7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.104
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.10
DNS Server
192.168.1.10

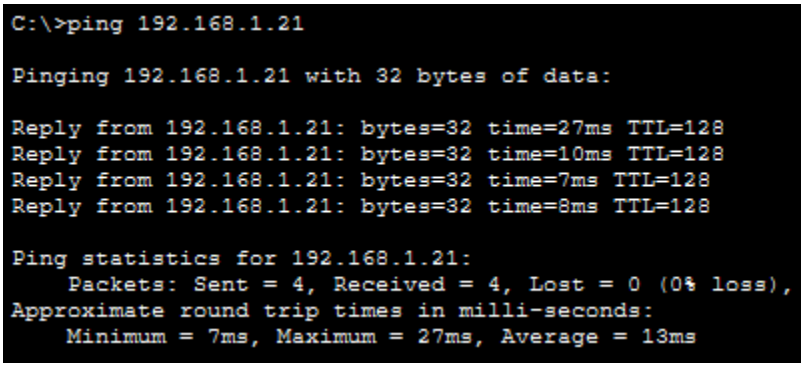

Client 8
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.105
Subnet Mask
255.255.255.0
Default Gateway
192.168.1.10
DNS Server
192.168.1.10

Paste Screenshots of ping replies from Client 5 to Client 1 & Client 2 here:

Client 5 to Client 1:

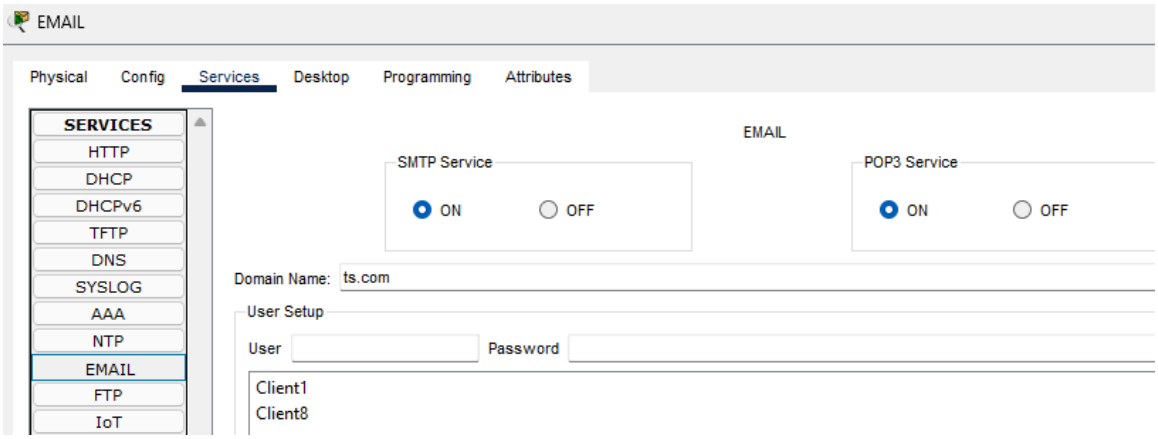


Client 5 to Client 2:

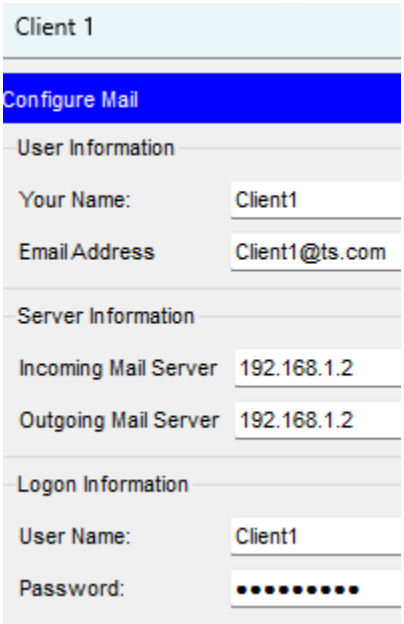


TASK 3 – Email Configuration

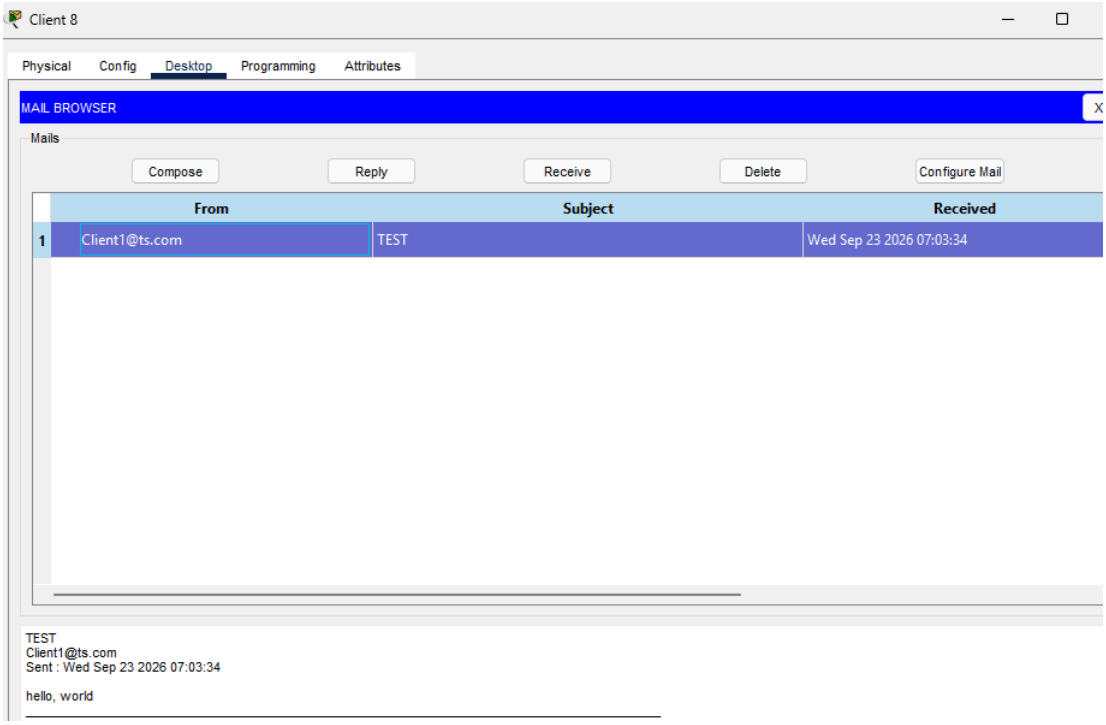
Paste Screenshot of EMAIL Server configuration here:



Paste Screenshot of Client 1 email configuration here:

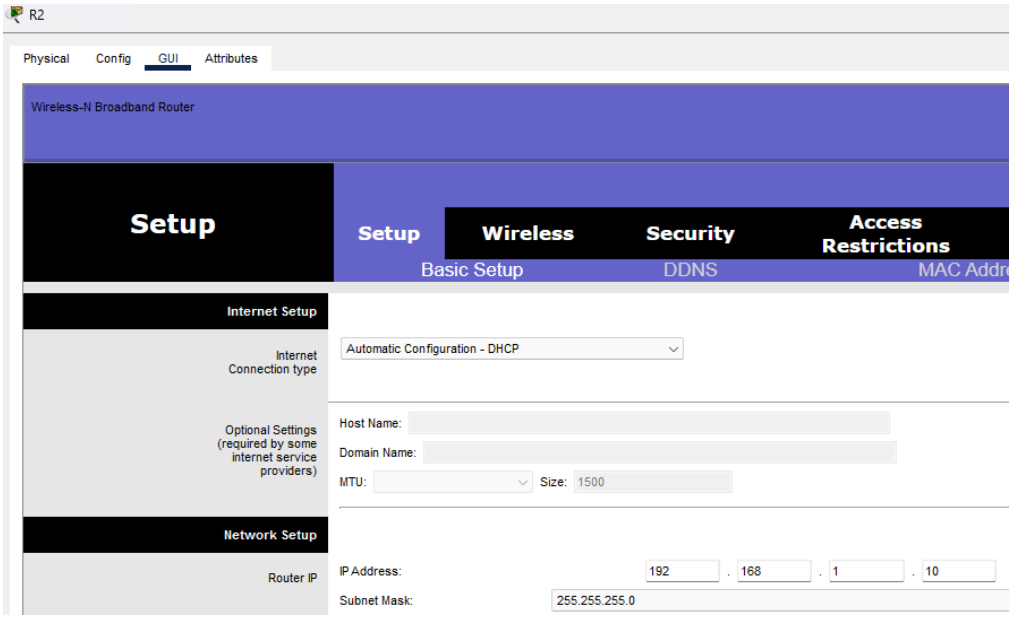


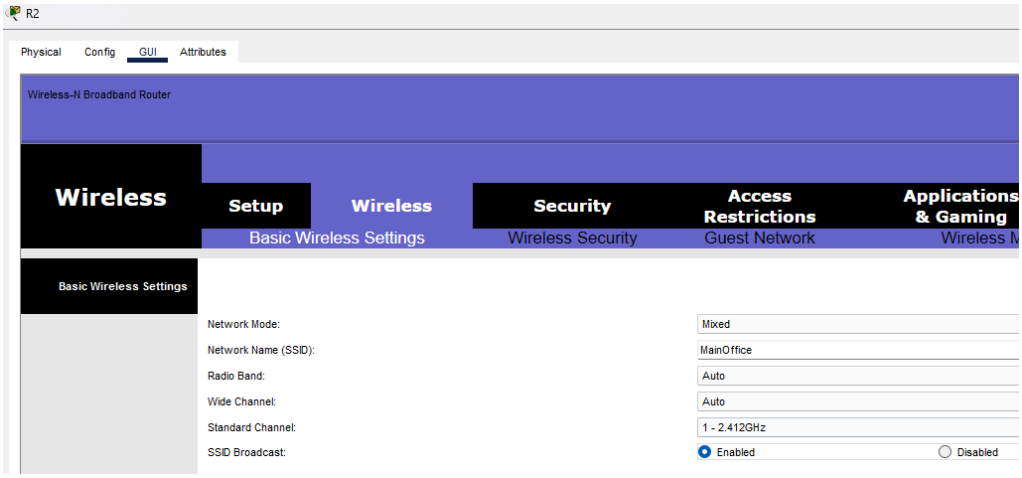
Paste Screenshot of received email on Client 8 here:



TASK 4 – Wireless Security Configuration

Paste Screenshot of wireless router basic settings configuration here:

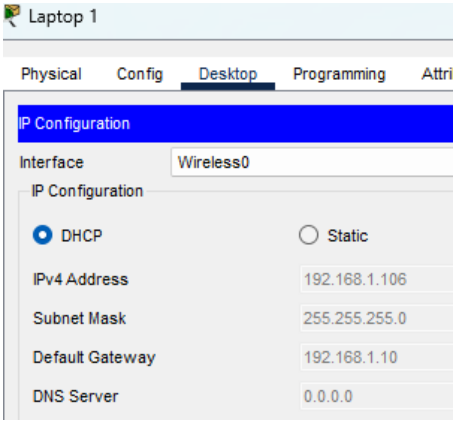




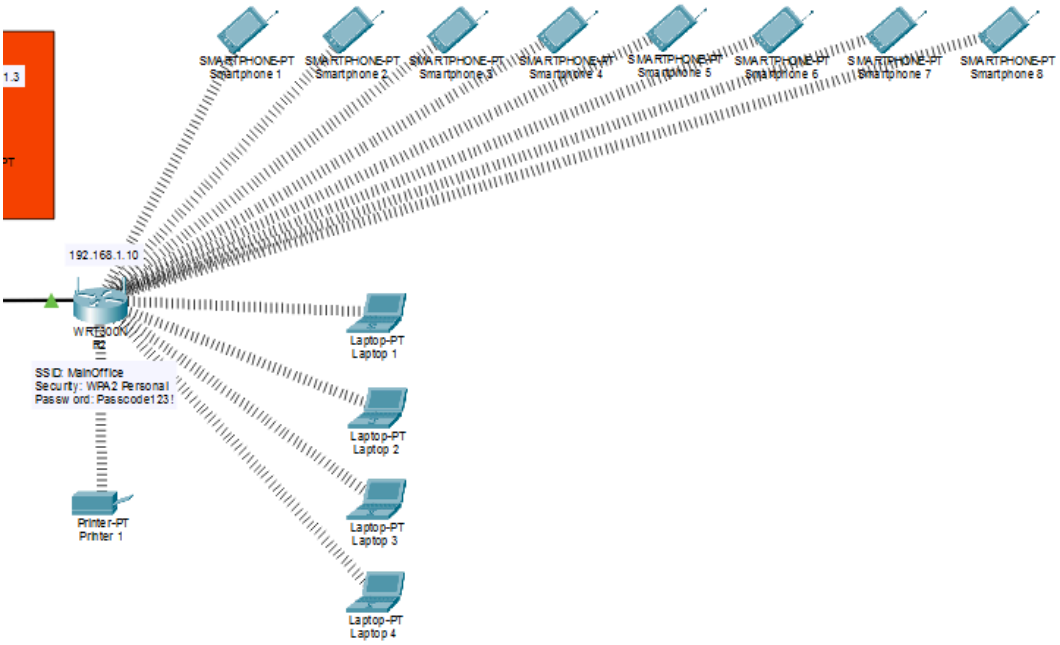
Paste Screenshot of wireless router security configuration here:



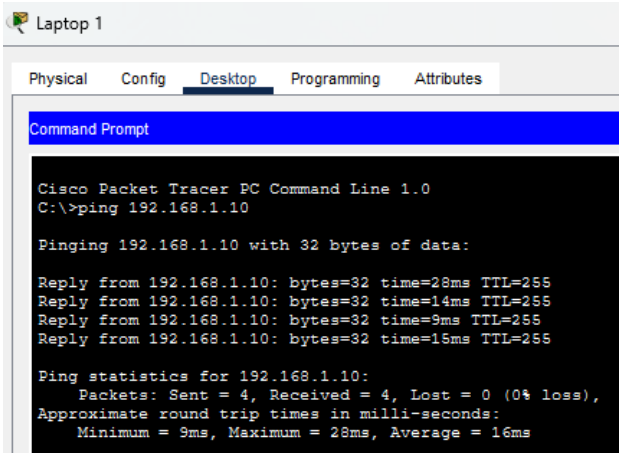
Paste Screenshot of Laptop1s wireless settings here:



Paste Screenshot showing all wireless clients connected here:

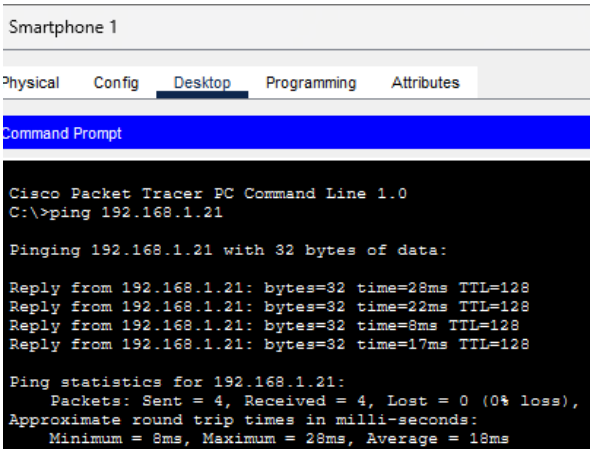


Paste Screenshot showing successful ping tests from any wirelessly connected laptop to gateway here:

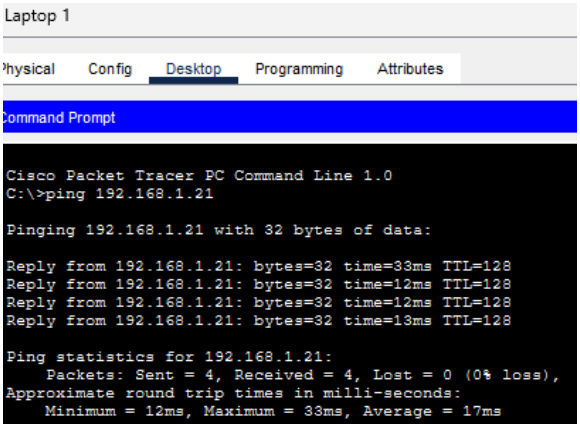


Paste Screenshot showing successful ping tests from Smartphone 1 to Client 1 and Laptop 1 to Client 1 here:

Smartphone 1 to Client 1:

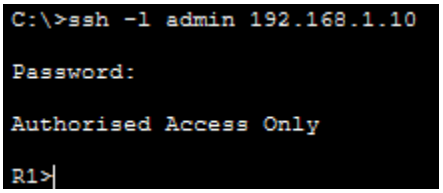


Laptop 1 to Client 1:



TASK 5 – Cybersecurity Hardening

Paste Screenshot showing successful SSH Login here:



Paste Screenshot showing MOTD and password request following R1 reload here:

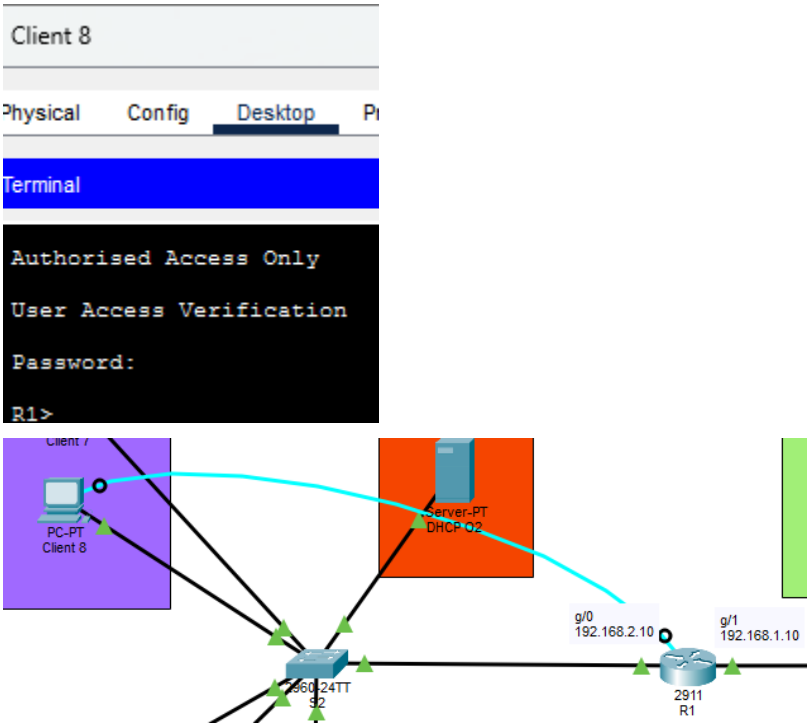
```
Press RETURN to get started!

Authorised Access Only

User Access Verification

Password:
```

Paste Screenshot of console cable connection between Client 8 and R1 here:



Paste Screenshot showing Running Config here:

```
R1#show running-config
Building configuration...

Current configuration : 934 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname R1
!
!
!
enable secret 5 $1$mERr$RldxcCZEZsTFTETUyRaA50
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
username admin password 7 0802455D0A16544541
!
!
license udi pid CISCO2911/K9 sn FTX152491LD-
!
!
!
!
!
!
!
!
!
!
!
ip domain-name ts.com
!
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 192.168.2.10 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.1.10 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
```

```
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^CAuthorised Access Only^C
!
!
!
!
line con 0
 password 7 0802455D0A16544541
 login
!
line aux 0
!
line vty 0 4
 login local
 transport input ssh
!
!
!
end
```

Final Submission Requirements

1. Packet Tracer File

Final working .pkt project (Annotated and Labelled correctly)

2. Evidence Document

Include:

- Screenshots
- Configuration evidence
- Testing evidence

Reflection / Conclusion

I created two separate office networks, connected them with a router, and configured each intermediary device and end device with the correct IP addresses, default gateways, subnet masks, and DNS servers according to the project specification. Some devices were assigned static IP addresses and configuration information, while others were dynamically allocated IP addresses and basic configuration information via DHCP pools configured on a DHCP server, with one server for each office.

Devices in the second office all had wired connections, while the main office consisted of devices with a combination of wired and wireless connections. To allow for wireless connectivity in the main office, I configured a wireless router with a unique SSID and WPA2-Personal security. Both the laptops and smartphones used this router to connect to the network without physical connections.

I then configured the email service on a dedicated email server located in the main office. To achieve this, I had to configure the domain specified in the project requirements and create the required email accounts. I then configured those email accounts on their respective hosts (Client 1 and Client 8) to allow those users to exchange emails via the email server. It was important that all these details matched the project specification; otherwise, the email accounts might not function correctly.

Once I finished configuring these devices and services, I began verifying their functionality through ping tests to check whether devices could communicate with one another. I also tested connectivity between the two offices, checked that devices received IP addresses through DHCP, and verified that wireless devices could connect to the network. I tested the email service by sending an email from Client 1 to Client 8 and confirming that it was received successfully.

Finally, I configured security settings on the router and switches, including passwords, password encryption, an MOTD banner, and SSH access on the router. This helped me understand the importance of securing network infrastructure and controlling access to network devices.

Overall, this project helped me develop practical skills in IP addressing, DHCP configuration, routing, wireless networking, email services, network security, and troubleshooting.